1 Response to the Referee: Neural Bellman Operators, R91

We thank the referee for distinguishing the scope of the original economic problem from the limited implications of its executed numerical catalogues. This revision responds to the report at `b030d87d7e96440eb25a3178c94ee6a87940cdb0`, which reviewed the R84 delivery. It starts from the intervening R90 manuscript at `9e06160b5464c41a5b84eaab9c3c4d171fc22832`, retaining the original title, author, economic primitives, substantive sections, broader applications and unfavorable evidence. The changes described below are amendments to that paper, not a proposal to replace it with a narrower numerical-method article.

The main new result is signed original-law qualification for complete multi-date policies under monotone cash-invariant recursive preferences. Theorem `thm:signed91` charges a selected-action upper discrepancy against an all-action lower discrepancy and retains the direction of numerical transition error. Proposition `prop:risk-transport91` gives explicit tail-risk and entropic transport coefficients. Full proofs, sharpness examples and counterexamples are included. The accompanying deterministic tests do not add independent economic observations.

The publication status is determined by the current verification manifest and a matching visual record. The manifest is named `VERIFICATION91.json`; it is stored in the reproduction-audit directory. This response does not convert a planned check, a compiler-produced PDF, or an inherited execution into current release approval.

1.1 M1 / B1. A complete, source-bound submission package

The R90 execution revealed concrete source-integration failures: a nested original-law module and the frozen R73 service data were absent from its sparse checkout, the R89 proposal module required by recording tests was missing, the recipe named a nonexistent `tests86.py`, active bibliography keys were duplicated, and required inherited tables were not reconstructed before typesetting. The new revision restores the R89 module by its exact historical Git blob and declares the full nested dependency paths in the publication checkout. The table reconstruction now includes the qualification-frontier and shared-readout tables as well as the newer economic and shared-proposal catalogues.

A newly authored 14-test suite supplies explicit regression examples for the inherited R86 joint-domain, acquisition, resource and price-region claims. It is identified as new R91 coverage, not a recovered R86 execution. The original bibliography files remain unchanged; the active catalogue removes duplicate citation keys only when their entries are identical or have the same DOI, retaining the full duplicate records. The journal class files are not modified. All inherited active section inputs remain in the main article, technical supplement or numerical appendix.

The current recipe runs every named current suite without skipped tests, the independent shared-readout checker and original-law cost replay, required table reconstruction, four-document typesetting and a separate clean typesetting comparison. It records actual failures. The current PDFs require subsequent source-matched rendered-page inspection. A successful build is evidence of delivery consistency, not mathematical correctness or editorial acceptance.

1.2 M2 / B2. Retaining the original problem while making its complete-policy content explicit

We have not adopted the recommendation to replace the paper with a first-date finite-menu article. The original NBO question concerns constructed continuations and feasible economic decisions, including complete policies and broader recursive objectives. The revision develops that question

rather than removing its difficult parts. The complete-policy occupation theorem, contingent-pattern disaggregation, joint state-context construction and stored-continuation resource account remain active.

The new signed theorem adds an explicit original-optimum comparison for nonlinear recursive evaluations. It permits changes in actions and continuations at all dates and uses uniform inequalities on the augmented economic state, not a fixed reference occupation or only the acquired states. Endogenous preference variables are included in that state. The scope does not assume that the already executed expectation experiment is an experiment on every nonlinear preference specification.

1.3 M3 / B2. Recursive propagation and the original objective

For each stored continuation, the revision verifies a lower original-minus-numerical action-value bound over every original feasible action and an upper bound for the implemented feasible action. Numerical residual span, greediness, the difference of those signed endpoints, and terminal span enter the original-policy loss certificate with their correct dated discounts. Recursive monotonicity and cash invariance establish the comparison with both the complete implemented policy and the original Bellman optimum. No linear expectation telescoping is asserted for nonlinear risk.

The formula contains the earlier symmetric transport allowance as a special case. A one-date singular-transition example attains the signed charge and shows a strict factor-two improvement over its symmetric counterpart. A two-date example shows why a common-action bias cannot be cancelled when it varies with the state: an earlier action changes which later bias is encountered. A separate four-outcome calculation distinguishes static and nested tail risk. The latter is used only when it is the original recursive primitive, not as a convenient replacement objective.

All feasible-action and own-future obligations remain explicit. The policy-specific upper endpoint does not license replacing the all-action lower endpoint by a selected-action check. The stationary discounted conclusion and fixed-opponent best-response interpretation retain their stated boundedness and original-objective conditions. Non-cash-invariant aggregators continue to use their separately proved absolute-error recursion.

1.4 M4 / B4. Joint-domain and prospective claims

The joint state-context construction and adaptive acquisition theorem remain in `population86`. Their guarantees require verified primitive regularity, feasible action transport and whole-domain coverage; finite catalogue agreement is not substituted for them. The conditional hit-probability condition permits dependent adaptive acquisition, but its lower bound must itself be justified. New exact regression examples test the affine-ReLU/min-plus identity on a joint domain, its label interpolation and a dependent acquisition process satisfying the stated hit bound.

The original held-out shared-proposal study remains separate. Its calibration and evaluation states, unseen dimension, fixed models and selector are retained. R91 does not refit after inspecting those outcomes. The new signed theorem is pointwise once its uniform premises are verified, but it does not turn the finite study into an unrestricted distribution-free generalization claim.

1.5 M5 / B3 / B6. Incremental neural work and complete resource accounting

All original query-work, complete-process timing, storage and diagnostic evidence remains available. The shared proposal's negative incremental ReLU query-work result is retained. The

calibration-selected conventional service and the matched polynomial comparator are not removed. Two architectures and repeated complete processes retain their original sample identities; replay is not additional independent evidence.

The new error formula changes the justified accuracy requirement, not an observed runtime. Local tail-risk or entropic sensitivity is explicitly paid in transition precision. The resource program continues to charge successor-circuit evaluation and the coupled dependence on later accuracy. A smaller resource upper bound is not treated as proof of a smaller runtime or as a comparison with another upper bound. Energy, application-specific monetary prices and a hardware causal treatment are not measured by this amendment.

1.6 M6 / B7. Original-law costs and uncertainty decomposition

The independent complete-policy investment evaluation, its prespecified margin, simultaneous inference allocation and full pairwise catalogue are retained. The selected controller's identified improvements and noninferiority statements remain distinct from the unresolved strict fine-basis ordering. Numerical enclosure, sampling and finite-support components remain visible; deterministic optimizer-class enlargement is not represented as additional sampling uncertainty.

The new fixtures concern exact finite-state recursive-risk values. They are used to challenge the mathematical error account, not to tighten the investment confidence intervals or increase their sample size. The original-law replay is separately invoked and source-bound. A new risk objective would require its own frozen economic evaluation rather than relabeling the expectation results.

1.7 M7 / B8. Economic prices and uniform decisions

The price-fee regions, capacity-constrained allocation analysis and all quality-only replacement-fee breakpoints remain in the manuscript and appendix, including negative ceilings. The robust region is a statement conditional on the supplied price set and simultaneous policy-cost enclosure. New regression examples check a whole price rectangle at its vertices and retain an unfavorable fee ceiling.

The tail-risk level in the new proposition is an original preference parameter, not an adjustable computational valuation chosen to obtain a favorable ranking. Approaching level one can make transition information more expensive: an exact two-atom example attains the required inverse-tail-mass factor. The manuscript does not invent an empirical monetary calibration to turn this mathematical sensitivity into a welfare estimate.

1.8 M8 / B5. Prospective selection

The fixed calibration/evaluation separation and all declared candidate services remain unchanged. The calibration-selected adaptive service had zero empirical query regret on the retained finite evaluation catalogues, including a tie; that fact is not asserted to be zero population regret or zero original-optimum policy loss. The independent complete-policy comparison has its own selected controller and evaluation stream.

A new regression example explicitly separates a calibration-selected service from the eventual test winner, so the convention cannot silently turn an ex post minimum into prospective evidence. The general prospective bounds remain conditional on their bounded-loss and sampling hypotheses.

1.9 M9 / B10. Horizon complexity and risk-adjusted precision

The stored-continuation account and coupled accuracy program remain active, together with the adverse vector descendant counts. The signed theorem extends their original-policy budget to the stated recursive risk primitives. Average value at risk contributes a local Lipschitz-to-transport factor divided by the surviving tail mass; entropic evaluation contributes an oscillation-dependent factor. Constant error is propagated through each cash-invariant recursion only by its date discount. It is therefore not multiplied by every later tail factor. This does not bound the continuation modulus independently of horizon: that modulus and its construction work remain charged.

The full matched large-vector tournament and an independent implementation of the largest vector recursion have not been executed by these exact diagnostics. They remain identifiable empirical obligations within the original scope. The theorem and new counterexamples are positive analytical advances, not substitutes for that larger computation.

1.10 M10 / B9. Prediction diagnostics and protected service

Earlier fitting warnings, invalid predictions, repairs, screen attempts and closures, prediction-free returns, refinements, zero-progress probes and adverse timings remain separated. Restoring the R89 module permits the original recording tests to exercise its invalid-predictor fallback rather than fail at import. A protected service return is not counted as successful prediction or as evidence of unrestricted fitting reliability.

R91 adds no training attempt. The earlier 212-service, 188-process, nine-controller and cross-machine catalogues have different units and are not aggregated into a favorable undifferentiated success rate. Every table retains its own source and denominator.

1.11 M11 / B11. Independent checking and exact coverage

The new numerical action-value implementation evaluates tail risk by minimizing over support thresholds. The separate original-law evaluator uses the capped-density dual and a distinct backward recursion, without importing that implementation or using a floating-point optimizer. All 64 complete three-date policies are checked at both initial states in each of 27 configurations. The resulting 1,728 policy evaluations and 3,456 inequalities are deterministic checks, not a population sample. Another 216 finite-law cases compare the two risk representations, retaining quantile atoms and zero-probability outcomes.

The 36 new signed-risk tests include invalid input rejection, exact ties, terminal and discount weights, constant-shift invariance, sharp transport factors, the failure of statewise bias cancellation and the difference between static and nested risk. The entropic transport assertion is proved analytically; this rational policy enumeration is not presented as an entropic experiment. The 14 additional tests of inherited R86 claims are separately identified. Existing independent original-law and common-readout checks retain their own scope. None is described as the still-unexecuted independent largest-vector solver.

1.12 M12 / B12. Manuscript organization and contribution

The reading path now connects the original construction, complete-policy qualification, joint-domain transfer, signed original-law recursive accuracy, coupled precision, prospective economic decisions and executed evidence. Full new proofs are in the technical supplement; all exact configurations are in the numerical appendix. Previous section inputs, original bibliography entries and adverse evidence are retained rather than deleted to shorten the argument artificially.

We do not claim priority for cash invariance, weak duality, the variational representation of average value at risk, Wasserstein coupling or geometric programming. The revision uses these principles to sharpen and extend the same NBO original-optimum account with explicit own-future, original-action, common-parameter and dated resource obligations. The signed sharpness and objective-preservation counterexamples clarify which improvements are justified. Whether these contributions meet the journal's novelty standard remains a matter for substantive review.

1.13 Minor comments and verification state

Exact catalogue winners remain qualified by their declared objective. Work, time, capacity, energy and monetary value are not conflated. Source, execution, reconstruction and delivery identities are distinct. Every active current bibliography key is unique while original entries are preserved. All new tables distinguish deterministic diagnostics from independent observations. Inherited vector results and warning catalogues retain their provenance.

Current completion is reported by the actual reproduction and visual records, not inferred from this response. A failed or unexecuted gate remains visibly failed or unexecuted. This submission preserves the original NBO paper, supplies the new full proofs and source-level repairs, and presents the resulting manuscript for another substantive referee assessment without replacing its topic or suppressing unfavorable evidence.